

Zin Win Thu, Ph.D.



Postdoctoral Researcher, Aerospace Engineering

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Research Experience

2026 – Present

■ **Postdoc Researcher**, Konkuk Aerospace Design and Airworthiness Institute (KADA), Konkuk University.

- Developing a deep generative and probabilistic surrogate modelling framework for uncertainty-based design optimization for aircraft design.
- Leading development of Comprehensive Aircraft Design Environment (CADE) in-house program, coordinating a team from M.S. students to research professors.
- Serving as design-team lead for design program development of the Medium UAV Common Platform (MUCP).

2020 – 2025

■ **Graduate Researcher**, Konkuk Aerospace Design and Airworthiness Institute (KADA), Konkuk University.

- Multidisciplinary aircraft design and analysis, including electric aircraft sizing and performance evaluation.
- AI-based design for 2D airfoils and 3D aircraft configurations, including uncertainty-based optimization.

Education

2020 – 2025

■ **M.Sc.–Ph.D. in Aerospace Engineering**, Konkuk University, Seoul, South Korea.
Dissertation: *GenAI Optimization Approach for Large Scale Aircraft Design under Uncertainty*.

2013 – 2018

■ **B.Sc. in Aerospace Engineering**, Myanmar Aerospace Engineering University, Myanmar.

Thesis: *Design and Construction of Mini Jet Engine*.

Publications

Peer-Reviewed Journal Articles

1. Min Ji Kim, Mohammad Irfan Alam, **Zin Win Thu**, Dong Oh Kwang, Jae-Woo Lee (2026). *Multi-step Hybrid Optimization Strategy with Adaptive Boundary Control Case Study: Advanced Air Vehicle Wing Design Optimization*. Aerospace Science and Technology, 111979.
2. **Zin Win Thu**, Maxim Tyan, Yong-Hyeon Choi, Mohammad Irfan Alam, Jae-Woo Lee (2025). *Possibility-Based Sizing Method for Hybrid Electric Aircraft*. IEEE Access.
3. Yong-Hyeon Choi, **Zin Win Thu**, Maxim Tyan, Woosuk Jung, Jae-Woo Lee (2025). *Balancing Power and Energy Needs in Hybrid Electric Aircraft Sizing: Payload and Technology Considerations*. Journal of the Korean Society for Aeronautical & Space Sciences, 53(4): 421–432.
4. Mohammad Irfan Alam, **Zin Win Thu**, Nghia Nguyen, Maxim Tyan, Jae-Woo Lee (2024). *Integrated Propulsion System Analysis Framework for Designing Advanced Air Mobility Aircraft*. IEEE Transactions on Transportation Electrification, 10(4): 10219–10238.
5. **Zin Win Thu**, Dasom Kim, Junseok Lee, Woon-Jae Won, Hyeon Jun Lee, Nan Lao Ywet, Aye Aye Maw, Jae-Woo Lee (2022). *Multivehicle Point-to-Point Network Problem Formulation for UAM Operation Management Used with Dynamic Scheduling*. Applied Sciences, 12(22): 11858.

6. **Zin Win Thu**, Aung Myo Thu, Phyo Wai Thaw (2018). *Design and Optimization of Aircraft Configuration for Minimum Drag*. International Journal of Aviation, Aeronautics, and Aerospace, 5(5), 10.
7. **Zin Win Thu**, Aye Aye Maw, Jae-Woo Lee (2025). *Deep Generative and Probabilistic Surrogate Modeling for Subsonic Airfoil Design*. Journal of Aircraft (under review).

Selected Conference Papers

1. **Zin Win Thu**, Min Ji Kim, Jae-Woo Lee (2026). *Efficient Aircraft Design through Deep Generative Modeling*. ACSMO 2026, Busan, Korea.
2. Yong-Hyeon Choi, **Zin Win Thu**, Dongoh Kwag, Jae-Woo Lee (2025). *Deriving the Flight Envelope and Conversion Corridor of an eVTOL Tiltrotor Aircraft Considering Battery Discharge Rate*. AIAA AVIATION FORUM AND ASCEND 2025, p. 3758.
3. DongOh Kwang, Min Ji Kim, **Zin Win Thu**, Jae-Woo Lee (2025). *A multi-objective optimization approach for reliable electric propulsion system architecture in AAM aircraft*. WCSMO16, Kobe, Japan.
4. **Zin Win Thu**, Jae-Hyun Ahn, Jae-Lyun Lee, Do-Youn Kwon, Yeon-Ju Choi, Woon-Jae Won, Maxim Tyan, Jae-Woo Lee (2022). *Enhanced Performance Prediction of Hydrogen Fuel Cell Powered eVTOL UAV*. AIAA AVIATION 2022 Forum, p. 3382.

Patents

- 2023  Jae-Woo Lee, **Zin Win Thu**, Dasom Kim, Aye Aye Maw. *UAM operation device and method based on a point-to-point network that guarantees dynamic scheduling*. Korea Patent No. 10-2023-0036070 (Granted).

Awards & Grants

- 2024  Grant for Outstanding Researcher Development Program, Konkuk University.

Teaching and Mentoring

- 2023 – 2024  **Teaching Assistant**, Comprehensive Aircraft Design, Konkuk University.
- Supported delivery of a senior-level aircraft design course (34 students).
 - Mentored projects on aircraft sizing, aerodynamics, propulsion, and performance analysis.
 - Delivered lectures on electric aircraft sizing and performance analysis.
-  **Mentorship**, Research for Undergraduate Students (RUS), Konkuk University.
- Mentored undergraduate researcher on electric regional aircraft sizing and power-train comparison, including sizing-program module development.

Volunteering Activities

- 2023 – Present  **IWIK Community Organization**, Volunteer English Teacher & Event Manager.
- Teach English to orphans and prepare weekly lessons.
 - Organize community events supporting elderly and low-income groups.
 - Managed major cultural and fundraising activities.

Research Project Participation

2022 – Present	Development of Comprehensive Aircraft Design Environment (CADE) in-house program.
2026 – Present	Design Program Development of Medium UAV Common Planform.
2023 – 2024	Concept Development of Next-Generation Regional Air Mobility Aircraft.
2021 – 2023	UAM Digital Twin.
2020 – 2023	Development of Small eVTOL UAV with Hydrogen Fuel Cell.
	Development of Personal Air Vehicle (PAV).

Skills

Programming	Python (advanced), MATLAB.
Machine Learning & AI	Generative Models: VAE. Probabilistic Models: Bayesian Neural Networks (BNN), Mixture Density Networks (MDN). Frameworks: TensorFlow, TensorFlow Probability. Applications: Surrogate modeling, uncertainty quantification.
Optimization & Uncertainty	Genetic Algorithms (GA, NSGA-II). Gradient-based optimization (SLSQP). Reliability-Based Design Optimization (RBDO). Robust Design Optimization (RDO).
Aerodynamics & Aircraft Design Tools	SU2, AVL, XFOIL, OpenVSP, SUAVE
Scientific & Technical Tools	LaTeX, Sphinx, MS Office.
Languages	English (fluent), Korean (basic), Burmese (native).

References

Available upon request